

Multiple Choice

1. **Answer:** C

Options: A) $O(\log e^N)$; B) $O(\log N)$; C) $O(\log \log N)$; D) $O(N)$

Note: Correct option: C - $O(\log \log N)$

2. **Answer:** D

Options: A) 0.1; B) 0.2; C) 0.3; D) 0.5

Note: Correct option: D - 0.5

3. **Answer:** A

Options: A) `**`; B) `//`; C) `is`; D) `not in`

Note: Correct option: A - `**`

4. **Answer:** C

Options: A) All of the preconditions in the program are correct.; B) All of the post-conditions in the program are correct.; C) The program may have bugs.; D) Every method in the program may safely be used in other programs.

Note: Correct option: C - The program may have bugs.

5. **Answer:** D

Options: A) The Caesar Cipher, a substitution cipher; B) DES (Data Encryption Standard), a symmetric-key algorithm; C) Enigma, a transposition cipher; D) One-time pad

Note: Correct option: D - One-time pad

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Merge sort'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'c'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $(a + b)*c + (a + b)*d$ '.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def Check_Solution(a,b,c): | return ("Yes") | return ("No")`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def is_sublist(l, s): | return sub_set`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key